Performance assessment of Ethernet 10BASE-T1S networks in automotive applications

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***Abstract*—The IEEE 802.3cg-2019 standard defines Ethernet 10BASE-T1S. With a data rate of 10 Mbps in half-duplex mode on a multidrop bus, Ethernet 10BASE-T1S is a suitable candidate for in-vehicle networks with zonal architectures to interconnect multiple sensors with the relevant zonal controller. The standard provides optional Physical Layer Collision Avoidance (PLCA) capabilities that support cyclic transmissions without collisions and provide short transmission delays. However, PLCA lacks explicit priority support, while priority is useful in automotive communications to handle different flows with diverse timing requirements. In this context, this work presents a simulative performance assessment of Ethernet 10BASE-T1S with PLCA enabled in a realistic zonal-based automotive scenario. Moreover, the paper addresses how to leverage the PLCA to offer a better Quality of Service (QoS) to the nodes that generate the most time-critical flows, discussing the delays obtained with different network configurations.**

***Index Terms*—Automotive Ethernet, Ethernet 10BASE-T1S, Real-time networks.**

I. Introduction

In-vehicle network architectures are changing from the traditional domain-based ones, in which the functions are grouped into logical domains leveraging the concept of functional domains [1] [2], to zonal architectures [3], in which the wiring of the electronic components does not depend on their functions, but on their physical location in the car.

In zonal architectures, the vehicle is divided into sections, called zones, each one managed by a zonal controller (ZC) that handles communications with nearby sensors and actuators and processes local data. Each ZC is connected with both the other ZCs and the central high-performance computer that performs complex tasks (e.g., decision-making for automated driving) through a high-speed backbone network, typically Ethernet-based. Zonal architectures not only allow for a clean network layout with fewer ECUs, but they also reduce wiring harness and packaging space, simplify installation and maintenance, and lower costs. In addition, zonal architectures facilitate data sharing and allow for flexibly adding new nodes (i.e., sensors, actuators), thus enabling the introduction of new functionalities throughout the vehicle lifecycle [4], [5]. Moreover, the transition towards Ethernet-based zonal architectures enables the adoption of multiple security protocols protecting the network against threats such as frame injection, replay attacks, etc. [6]. While high-speed automotive Ethernet networks running the Time-Sensitive Networking (TSN) family of standards are the de-facto candidates for the backbone technology [7], the communication between sensors/actuators and zonal controllers is not yet tied to a specific communication technology. However, moving towards a homogeneous network, entirely Ethernet-based, would simplify addressing and data transmission between nodes and avoid the need for complex gateways to interconnect different protocols, such as CAN FD and 100BASE-T1. For this reason, it is worth investigating the adoption inside a zone of the novel Ethernet 10BASE-T1S technology defined in the IEEE 802.3cg-2019 standard [8] that operates at 10Mbps over a single-pair Ethernet multidrop bus. The standard [8] defines a Reconciliation Sublayer that provides optional Physical Layer Collision Avoidance (PLCA) capabilities among participating nodes. The PLCA RS is specified for PHYs operating in half-duplex multidrop mode and enables bandwidth-efficient and collision-free transmissions. When PLCA is enabled, each node cyclically has one dedicated transmission opportunity (TO) in which it can transmit without collisions either a single frame or a configured number of frames if the burst mode is enabled. Although Ethernet 10BASE-T1S offers bounded transmission cycles, it does not support flow prioritization, as it was designed for simple devices that do not support the traffic classes defined in the IEEE 802.1Q standard [9] for end-nodes. The aim of this work is to assess the performance of time-critical flows over Ethernet 10BASE-T1S when PLCA is enabled and investigate how to leverage the PLCA to offer a better Quality of Service (QoS) to the nodes that generate the most time-critical flows. The contribution of this paper is the evaluation of the end-to-end delays obtained through OMNeT++ simulations in a realistic zonal-based automotive scenario. In particular, the paper evaluates two configurations, which differ in the number of transmission opportunities provided to the nodes. The first configuration assigns one transmission opportunity per cycle to each node, while the second one grants multiple transmission opportunities per cycle to the nodes that generate the flows with the most stringent time requirements.

The paper is structured as follows. Section II discusses related works. Section III describes Ethernet 10BASE-T1S with PLCA enabled. Section IV presents the assessed configurations and the simulation results. Section V discusses some limitations of the real-time behavior of PLCA. Finally, Section VI gives conclusions and hints for future work.

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II. Related Work

Recent works in the literature have focused on the use of Ethernet 10BASE-T1S for automotive applications. In [10], a comparative assessment between CAN, CAN FD, FlexRay, and Ethernet 10BASE-T1S networks is presented that evaluates the maximum transmission cycle needed to transmit all the network messages. The evaluation does not consider realistic scenarios, featuring, for example, periodic messages generated with different periods and characterized by diverse timing requirements. The work in [11] shows some insights on the performance of Ethernet 10BASE-T1S with different configuration parameters. However, the presentation merely shows some results without providing the simulated scenario and parameters. In contrast, this work presents an assessment of Ethernet 10BASE-T1S with PLCA enabled in a realistic zonal-based automotive scenario, with two well-specified network configurations.

As Ethernet 10BASE-T1S does not support node prioritization, when PLCA is enabled nodes transmit cyclically during their assigned TOs only. The works in [12], [13] and [14] proposed two strategies to prioritize nodes and reduce the maximum latency of the flows transmitted by high-priority nodes. In both strategies a distinction is made between low-priority and high-priority nodes, and each node is assigned a TO. However, while the low-priority nodes can transmit only in their assigned TOs, the high-priority nodes, in addition to this, can also interrupt the PLCA cycle by transmitting their frames outside their TOs.

In particular, in [12] the high-priority nodes can send their frames immediately after the transmission of any other node. Each high-priority node is assigned a different priority level and the MAC interpacket gap (IPG) is divided into n priority slots that correspond to the different priority levels assigned to the high-priority nodes. A high-priority node with a frame to send will send a signal, called a transmission request, during its dedicated priority slot, provided that the previous priority slots, which are assigned to the nodes with higher priority, have not been used. Consequently, each IPG can be followed by the transmission of at most one high-priority frame. In the case all n high-priority nodes have a frame to send, the mechanism is repeated n times. Upon reception of the transmission request signal, all the nodes stop counting the sequence of TOs. The regular PLCA cycle resumes as soon as there is an IPG in which no priority slot is used. Moreover, to further reduce the delay due to an ongoing low-priority transmission, the approach in [12] entrusts the PLCA with frame fragmentation for low-priority nodes. During transmission, if the low-priority frame size exceeds a configured threshold, the PLCA splits it into multiple fragments that are sent one by one during the node TO. On the receiving side, the PLCA buffers all the fragments until all of them have been received and then reassembles the low-priority frame before forwarding it to the MAC layer, which is unaware of the fragmentation.

The approach proposed in [13] and [14] is an enhancement of the one in [12] that allows a high-priority node with a frame ready to be transmitted to send a signal, called an interrupt. If the channel is free, the high-priority node sends the interrupt signal followed by the frame, otherwise, the high-priority node can send the interrupt only if the ongoing transmission on the channel has originated from a node with a lower priority. In this case, all the nodes detect a collision on the channel. The high-priority nodes wait for a shorter IPG than the value provided in the Ethernet standard, so they send their frames earlier than the low-priority ones, which instead wait for the regular IPG duration. Moreover, each high-priority node waits for a different IPG depending on its priority, i.e., the higher the priority, the shorter the IPG, thus enforcing a transmission order between the high-priority nodes. Upon receiving the interrupt, all nodes stop counting the TOs sequence. The PLCA resumes once the high-priority node that issued the interrupt stops transmitting, provided that no other high-priority nodes have frame to transmit.

While the strategies proposed in [12], [13], [14] entail modifications to the IEEE 802.3cg-2019 standard and are not compliant to it, the strategy adopted in this work prioritizes the most time-critical flows by leveraging on suitable configurations of the PLCA without modifying it, and therefore is fully compliant with the above mentioned standard.

III. Background

The Ethernet 10BASE-T1S technology introduced in the IEEE 802.3cg-2019 [8] amendment to the IEEE 802.3-2018 standard is now enrolled into the IEEE 802.3-2022 standard [15]. This standard defines a set of Ethernet physical layers operating at 10Mbps that encompass both point-to-point links and multidrop bus topologies on a single twisted-pair cable. Point-to-point links can operate in either half-duplex or full-duplex mode. The multidrop bus topology works only in half-duplex mode and supports at least up to eight nodes connected to a bus with a maximum length of 25 meters.

The architecture of an Ethernet 10BASE-T1S node with PLCA enabled is illustrated in Fig. 1. The MAC layer runs the classical CSMA/CD, while the PLCA Reconciliation Sublayer (RS) offers the PLCA capabilities. The PLCA RS interfaces with the Physical Coding Sublayer (PCS) via the Media Independent Interface (MII). The PCS handles 4B/5B encoding, translating 4-bit data into 5-bit symbols, and forwards them to the Physical Medium Attachment (PMA), which modulates the signals for transmission and prepares the received signals for decoding. Finally, the PLCA RS maps the MII signals to the Physical Signaling Sublayer (PLS) service primitives used by the MAC. PLCA allows nodes to transmit cyclically without collisions, thus obtaining more predictable transmission times, by assigning each network node one or more IDs that are used to schedule the transmissions. Each ID is assigned to only one node and corresponds to a single TO within the PLCA. As shown in Fig. 2, time is divided into cycles, each initiated by a Beacon signal broadcast to all the network nodes by the coordinator node, which is identified with the ID 0. After the Beacon, the cycle is divided into time-slots, called transmission opportunities (TOs), with one TO allocated

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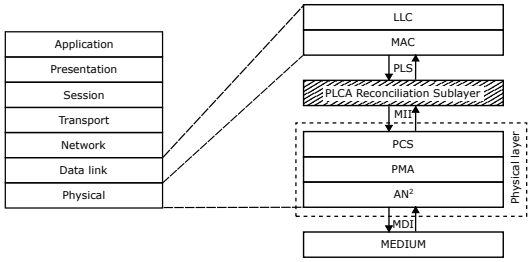

Fig. 1. Ethernet 10BASE-T1S node architecture with PLCA enabled.

to each ID. This way, the node with ID 0 is assigned TO 0, the node with ID 1 is assigned TO 1 and so on. Each node, as soon as the bus is idle, starts counting the sequence of TOs. At the beginning of its TO, each node is allowed to transmit a single frame. If there is no frame available, the TO is yielded (e.g., Node 2 in Fig. 2), otherwise, the node sends a Commit signal followed by the frame (e.g., Node 1 in Fig. 2). After receiving the Commit signal, the other nodes stop counting the TOs and only resume when the bus becomes idle. If the TO is yielded, it has a minimal duration (a few bit times), otherwise, if a frame is transmitted, the TO duration increases to accommodate the entire frame transmission. At the end of the frame transmission, the node sends two more signals, i.e., ESD followed by ESDOK in case of no transmit error (e.g., Node 1 in Fig. 2), which indicate the end of the TO. Optionally, a burst mode can be enabled that gives a node the ability to transmit more than one frame in a single TO. However, the burst-enabled mode allows only consecutive transmissions from the same node in a single TO. Here, instead, we address the case for assigning multiple TOs to a node on the basis of the temporal requirements of the frames that it needs to transmit. This way, the node will be allowed to transmit during the next assigned TO within the same PLCA cycle, instead of waiting for its single TO in the next PLCA cycle.

According to the IEEE 802.3cg-2019 standard [8] the MAC layer is not aware of the PLCA, which is implemented at the physical layer. For this reason, at any time it may happen that the MAC layer starts a transmission outside the TO assigned to the node. For instance, in Fig. 3, the MAC layer of Node 3 delivers its frame to the PLCA RS layer at time t_0, but then the PLCA RS buffers the frame until the Node 3 TO, i.e., time t_3. While the PLCA RS is waiting for the Node 3 TO, it may happen that another node (i.e., Node 1 in Fig.3) sends its frame over the physical channel (time t_1 in Fig. 3). In this case, the PLCA reports to the Node 3 MAC layer a collision (actually, it is a logical collision, as the Node 3 frame was not sent over the physical channel). This way, the MAC layer stops transmitting the frame and enters the first backoff period. As the first backoff period duration is shorter

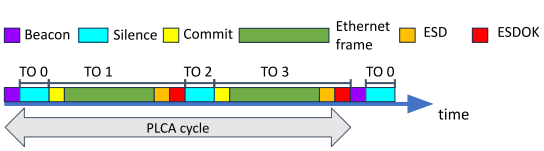

Fig. 2. Example of PLCA cycle.

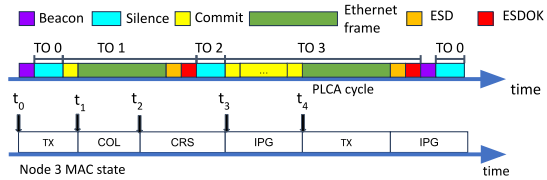

Fig. 3. Example of logical collision.

than the transmission time of the shortest Ethernet frame, the PLCA prevents Node 3 MAC layer from transmitting up to its next TO by keeping high the Carrier Sense (CRS) signal to the MAC layer (from time t_2 to time t_3 in Fig. 3). When the Node 3 TO arrives (time t_3 in Fig. 3), the PLCA lowers the CRS signal to allow the MAC layer to start transmitting after one IPG. While the MAC layer is waiting for the IPG to elapse, Node 3 sends a sequence of Commit signals to hold its TO. After the IPG, the frame is transmitted on the bus (time t_4 in Fig. 3).

The PLCA cycle has a variable duration. If all nodes yield their TO, the cycle has the shortest duration. In contrast, if all nodes transmit frames, the cycle has the largest duration. If there are N TOs, the minimum duration of the PLCA cycle is calculated as in Formula (1),

$$T_{cycle,min} = 1.25 \times \frac{L_{Beacon} + N \times L_{TO}}{R_b} \tag{1}$$

where L_{Beacon} is the Beacon signal length in bits and is fixed in the standard [8] to 20 bits. L_{TO} is the TO length in bits and is a configuration parameter (its default value is 32 bits). R_b is the data rate of the physical channel, i.e., 10Mbps. As PLCA applies a 4B/5B coding, i.e., coding 4 bits into 5 bits, the cycle bit length is multiplied by $5/4 = 1.25$. The maximum duration of the PLCA cycle is calculated as in Formula (2),

$$T_{cycle,max} = 1.25 \times \frac{L_{Beacon} + \sum_{i=1}^{N}(L_{i,max})}{R_b} \tag{2}$$

where $L_{i,max}$ is the maximum transmission length in bits of the i-th TO plus the overhead. $L_{i,max}$ is calculated as in Formula (3),

$$\begin{aligned} L_{i,max} = L_{IPG} + L_{Commit} + L_{ETH} + L_{data,i,max} + \\ + L_{ESD} + L_{ESDOK}, \end{aligned} \tag{3}$$

where L_{IPG} is the IPG length in bits and is fixed to 96 bits. L_{Commit} is the Commit signal length in bits and is fixed in the standard [8] to 4 bits. L_{ETH} is the 26 bytes Ethernet overhead (i.e., 208 bits), which includes the frame preamble (8 bytes), Ethernet header (14 bytes) and frame check sequence (4 bytes). $L_{data,i,max}$ is the maximum frame size in bits that the Node i can transmit. L_{ESD} and L_{ESDOK} are the lengths in bits of the ESD and ESDOK signals, both fixed in the standard [8] to 4 bits.

IV. Simulative Assessment

The simulations aim to evaluate the Ethernet 10BASE-T1S PLCA performance in a realistic zonal-based automotive

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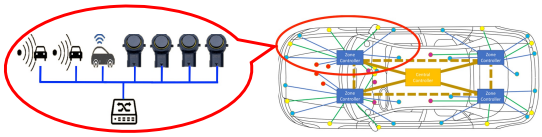

Fig. 4. Simulated network.

TABLE I
TRAFFIC FLOW CONFIGURATION.

Flows	Flow count	Frame Payload (B)	Period (ms)	D (ms)
Radar Data	2	625	4	4
Ultrasonics Data	4	188	100	100
Telematics Data	1	1500	6	6
HeartBeat	7	64	100	100

scenario in which the nodes transmit multiple flows, with different generation periods and temporal constraints, under two network configurations. The simulative assessment was performed using OMNeT++. The simulation model of Ethernet 10BASE-T1S physical layer was developed from scratch, while the INET framework models were adopted for the MAC layer and the upper layers. As Ethernet 10BASE-T1S is well-suited for zonal architectures the simulated scenario includes a zone, as shown in Fig. 4. The network consists of 8 nodes connected to a bus. The simulated network includes one zonal controller (ZC) acting as an Ethernet 10BASE-T1S coordinator, and seven end-nodes, i.e., two radars, one telematics and four ultrasonics. Each end-node transmits two flows, i.e., a Data flow and a HeartBeat flow, so there are 14 flows in total. The relative deadlines of the flows are assumed to be equal to their generation periods, which is realistic in automotive applications. The parameters of the flows, shown in Table I, are inspired by the realistic automotive scenario presented in [7].

The simulations consider very simple end-nodes that do not implement any of the QoS policies defined in the IEEE 802.1Q standard [9]. For this reason, the two flows generated by each end-node are transmitted in a FIFO order. All the frames are transmitted from the end-nodes to the zonal controller.

A. Configurations

Two configurations were simulated. In the first one, hereinafter called Single ID, each node is assigned a single ID (i.e. one TO) for each PLCA cycle. This configuration does not provide any QoS differentiation for time-critical flows. Conversely, the second configuration, hereinafter called Multiple IDs, enables QoS differentiation by assigning multiple IDs (i.e., multiple TOs) within a single PLCA cycle to the nodes that generate the flows with the most stringent requirements.

The PLCA ID assignment is shown in Table II. In the Single ID case, the IDs are assigned sequentially. In this case, the ID value assigned to a node does not affect the delays, as the applications are not synchronized with the PLCA. In fact, for example, if the node with ID 1 generates a frame right after the TO corresponding to ID 1 has passed, the node will have to wait for the relevant TO in the next PLCA cycle before transmitting. In the Multiple IDs configuration, as a first step, each node is assigned an ID sequentially. In the second step, based on the PLCA duration, calculated as in Formula (3), the nodes that may experience deadline miss are identified. As a third step, each of these nodes is assigned an additional ID. In the case here reported, the additional ID is chosen so that the TOs assigned to each node are temporally equally spaced. The second step is repeated until no flows that may experience deadline miss are found.

The TO duration is set to 32 bit time, which is the default value in the standard [8]. The burst mode was disabled in both configurations to prevent a node with multiple frames to transmit from delaying the transmissions of the nodes with higher IDs. In the assessed scenario, this situation would entail missed deadline due to the too large PLCA cycle.

TABLE II
PLCA CONFIGURATION.

Node	PLCA ID	
	Single ID	Multiple IDs
Coordinator	0	0
Radar 1	1	[1, 4, 7, 14]
Radar 2	2	[2, 5, 8, 15]
Telematics	3	[3, 6]
Ultrasonics 1	4	[9, 13]
Ultrasonics 2	5	[10, 16]
Ultrasonics 3	6	[11, 17]
Ultrasonics 4	7	[12, 18]

B. Simulation results

The performance metrics here considered is the end-to-end delay ($E2EDelay$) defined as in Formula (4), i.e., as the time interval between the frame generation at the source node ($GenTime$) and its complete delivery at the destination node, i.e., the zonal controller ($RxTime$), measured at the application level.

$$E2EDelay = RxTime - GenTime \tag{4}$$

Table III shows the average and maximum delays obtained in the case that all the flows start at the same time (i.e., they have the same offset). This is an unfavorable case, as all flows interfere with each other. Moreover, the Table shows the comparison between the Single ID configuration and the Multiple IDs configuration.

For the Single ID configuration, the Radar Data 1 flow obtained a maximum E2EDelay equal to 4.509 ms, which resulted in a deadline miss, while in the Multiple IDs configuration no deadline miss was experienced for any Radar Data flow. In the case of the Radar Data 2 flow, no deadline miss was experienced in either configuration. Since each node handles two flows (i.e., the Data and HeartBeat flow) that in this simulation start at the same time, both frames are ready to be sent at the same time and it might happen that the MAC layer transmits first the HeartBeat and then the Data flow. With this transmission order, in the Single ID configuration the Data frame is sent after two PLCA cycles (being the first cycle used for HeartBeat and the second one for Data). The maximum PLCA cycle duration in the Single ID configuration, calculated as in Formula (2), is equal to 3.78 ms and therefore the duration of two PLCA cycles is equal to 7.56 ms. This value is higher than the Radar Data relative deadline, which is 4 ms. As expected, the Multiple IDs

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TABLE III
END-TO-END DELAY SIMULATION RESULTS CASE 1.

Flow	D (ms)	Mean (ms)		Max (ms)	
		Single ID	Multiple IDs	Single ID	Multiple IDs
Radar Data 1	4	0.933 ± 0.006	0.960 ± 0.006	4.509	3.170
Radar HeartBeat 1	100	0.452 ± 0.021	0.324 ± 0.015	3.880	2.963
Radar Data 2	4	1.380 ± 0.005	1.310 ± 0.005	2.973	3.170
Radar HeartBeat 2	100	0.437 ± 0.018	0.288 ± 0.011	3.984	3.067
Ultrasonics Data 1	100	0.630 ± 0.014	1.784 ± 0.059	3.087	4.092
Ultrasonics HeartBeat 1	100	3.027 ± 0.050	2.706 ± 0.053	4.507	4.507
Ultrasonics Data 2	100	0.760 ± 0.013	1.889 ± 0.060	3.315	4.092
Ultrasonics HeartBeat 2	100	2.941 ± 0.053	3.403 ± 0.064	4.509	5.836
Ultrasonics Data 3	100	0.844 ± 0.015	1.982 ± 0.061	3.543	4.096
Ultrasonics HeartBeat 3	100	2.750 ± 0.056	3.471 ± 0.063	4.509	5.839
Ultrasonics Data 4	100	0.918 ± 0.019	2.063 ± 0.065	3.770	4.095
Ultrasonics HeartBeat 4	100	2.535 ± 0.056	3.429 ± 0.067	4.509	5.839
Telematics Data	6	2.135 ± 0.008	2.129 ± 0.008	4.403	4.658
Telematics HeartBeat	100	0.469 ± 0.016	0.841 ± 0.053	4.088	3.171

configuration meets the deadline requirements of the flows. This demonstrates that the Multiple IDs configuration makes it possible to reduce the delays of the time-critical flows that in the Single ID configuration experienced deadline miss, so that they meet their relative deadlines. The Telematics Data flow did not experience any deadline miss in either configuration. In fact, the maximum E2EDelay is lower than 6 ms, i.e., the relative deadline of the flow. The Ultrasonics Data flows have a very large relative deadline, i.e., 100 ms. In the Multiple IDs configuration, the Ultrasonics node has two TOs per cycle and therefore, if there are both Data and HeartBeat frames ready to be sent, each frame is sent in one of the two assigned TOs. As in the Multiple IDs case the PLCA cycle duration is longer than in the Single ID configuration, the E2EDelays are higher than in the Single ID case, but they are always lower than the relative deadlines of the flows. Finally, the HeartBeat flows did not experience any deadline miss in this scenario.

Simulations were also performed assuming that the flows do not start simultaneously. This is a less pessimistic case than the previous one, as the offset shift between the periodic flows limits their mutual interference. Shorter PLCA cycles and lower delays than in the previous simulations were obtained by both the Single ID and Multiple IDs configurations. The results are not shown here due to space limitations.

V. CONSIDERATIONS ON THE PLCA TIMINGS

In Ethernet 10BASE-T1S the application that generates the messages is not synchronized with the PLCA, the MAC layer is not aware of the PLCA and works in half-duplex mode. These aspects may occasionally lead a node to skip the transmission opportunity although the MAC layer has data to transmit. For instance, as shown for Node 2 in Fig. 5, it may happen that, while Node 2 is receiving a frame from Node 1, a new frame arrives at the Node 2 MAC layer at time t_1. Since the channel is currently busy (as indicated by the CRS signal), the Node 2 MAC layer cannot immediately transmit the new frame, but, before trying the transmission, it must wait first for the ongoing reception to complete, then for the channel to become idle, and finally for an IPG. Being unaware of the pending frame in the MAC layer until time t_3, the Node 2 PLCA fails to send a Commit signal to hold its TO at time t_2, thus skipping the transmission opportunity and, consequently, it will transmit the frame at the next TO, thus increasing latency.

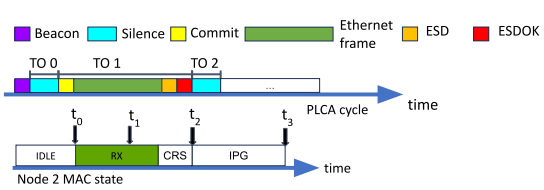

Fig. 5. Frame enqueued during a frame reception.

VI. CONCLUSION

The simulation results here presented demonstrated that, in Ethernet 10BASE-T1S with PLCA enabled, granting multiple TOs within one cycle to a node allows to meet the deadlines of the flows generated by the node. Future work will investigate a methodology for assigning the node IDs to implement fine-grained deadline-aware scheduling and the adoption of security mechanisms, leveraging the ongoing work within the Open Alliance TC17 on the adoption of the MACsec protocol on Ethernet 10BASE-T1S networks.

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